



ASSIGNMENT No: 2

TITLE: Develop a Java application demonstrating the use of default, parameterized, and copy constructors for object initialization and data management.

Problem Statement

Develop a Java application demonstrating the use of default, parameterized, and copy constructors for object initialization and data management.

Learning Objectives

To understand and implement default, parameterized, and copy constructors for object initialization.

Theory

Constructors are special member functions that are automatically invoked when an object is created. Java supports default constructors provided by the compiler, parameterized constructors for initializing objects with user-defined values, and copy constructors (implemented manually) for creating duplicate objects. Constructors improve code readability and simplify object initialization while supporting object-oriented programming principles.

1. Default Constructor

Code

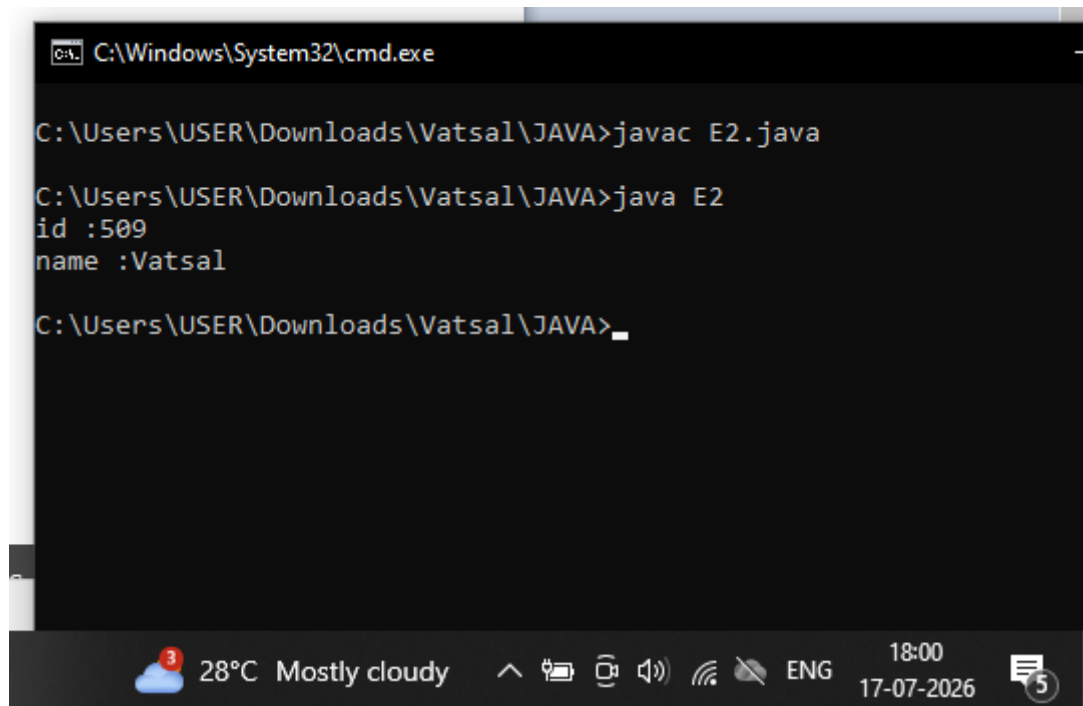
```
public class E2 {
    int id;
    String name;

    E2() {
        id = 509;
        name = "Vatsal";
    }

    void display() {
        System.out.println("id:" + id);
        System.out.println("name:" + name);
    }

    public static void main(String[] args) {
        E2 e = new E2();
        e.display();
    }
}
```

Output



```
C:\Windows\System32\cmd.exe

C:\Users\USER\Downloads\Vatsal\JAVA>javac E2.java

C:\Users\USER\Downloads\Vatsal\JAVA>java E2
id :509
name :Vatsal

C:\Users\USER\Downloads\Vatsal\JAVA>_
```

2. Parameterized Constructor

Code

```
public class Ex2 {
    int id;
    String name;

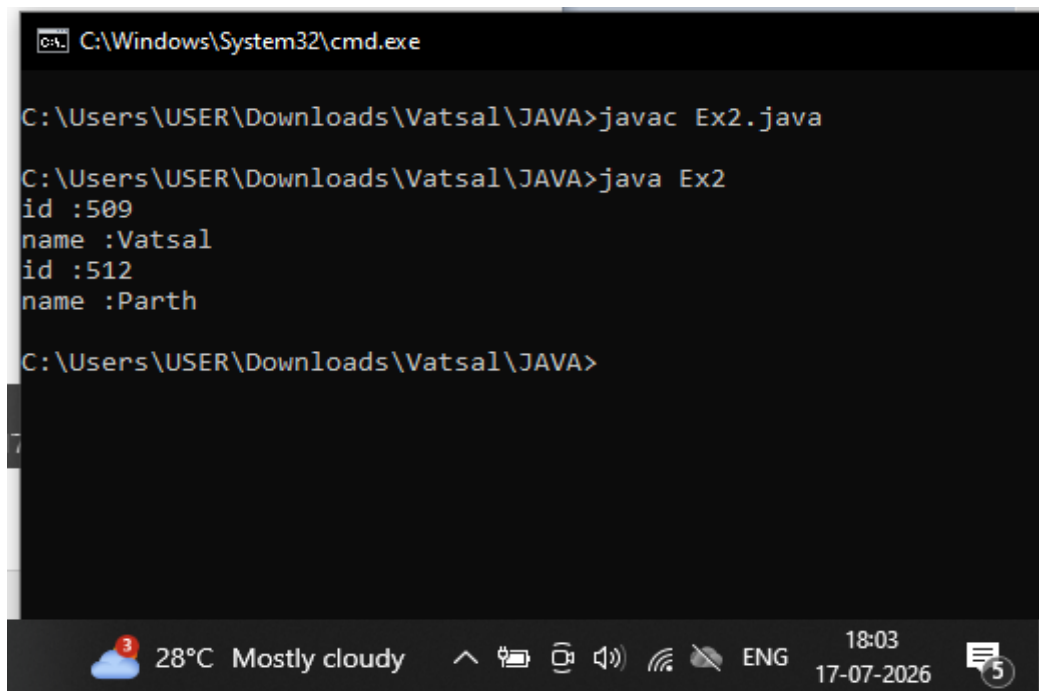
    Ex2(int i, String n) {
        id = i;
        name = n;
    }

    void display() {
        System.out.println("id:" + id);
        System.out.println("name:" + name);
    }

    public static void main(String[] args) {
        Ex2 e1 = new Ex2(509, "Vatsal");
        e1.display();

        Ex2 e2 = new Ex2(512, "Parth");
        e2.display();
    }
}
```

Output



```
C:\Windows\System32\cmd.exe

C:\Users\USER\Downloads\Vatsal\JAVA>javac Ex2.java

C:\Users\USER\Downloads\Vatsal\JAVA>java Ex2
id :509
name :Vatsal
id :512
name :Parth

C:\Users\USER\Downloads\Vatsal\JAVA>
```

3. Copy Constructor

Code

```
public class Exp2 {
    int id;
    String name;

    Exp2(int i, String n) {
        id = i;
        name = n;
    }

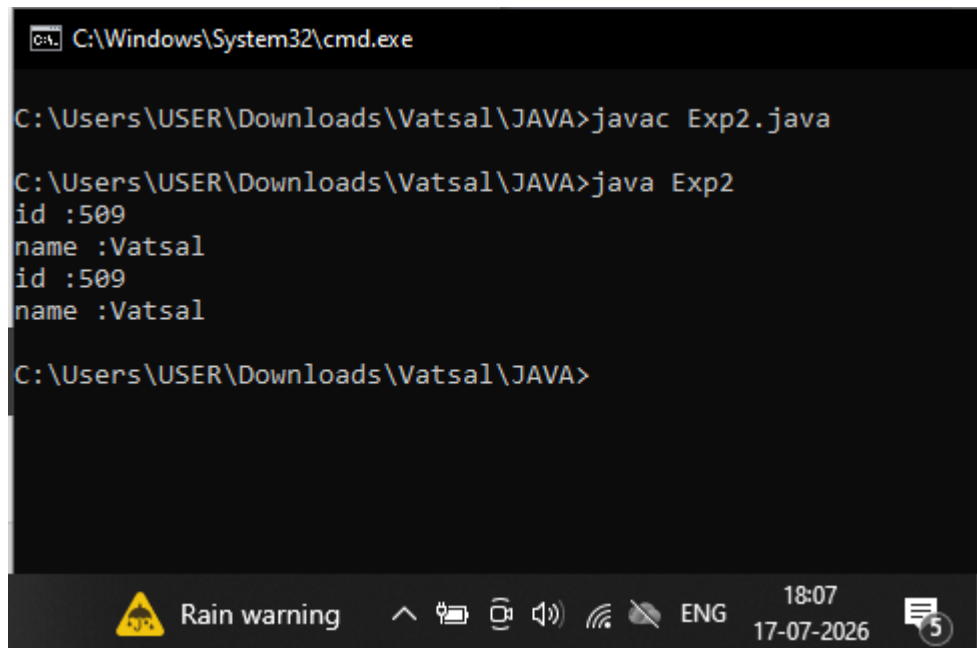
    Exp2(Exp2 s) {
        id = s.id;
        name = s.name;
    }

    void display() {
        System.out.println("id:" + id);
        System.out.println("name:" + name);
    }

    public static void main(String[] args) {
        Exp2 e1 = new Exp2(509, "Vatsal");
        e1.display();

        Exp2 e2 = new Exp2(e1);
        e2.display();
    }
}
```

Output



```
C:\Windows\System32\cmd.exe

C:\Users\USER\Downloads\Vatsal\JAVA>javac Exp2.java

C:\Users\USER\Downloads\Vatsal\JAVA>java Exp2
id :509
name :Vatsal
id :509
name :Vatsal

C:\Users\USER\Downloads\Vatsal\JAVA>
```

Conclusion

In this program, we learned how different types of constructors work in Java. We understood the use of default, parameterized, and copy constructors to create and initialize objects in different ways. A constructor in Java is a special type of method that is automatically called when an object of a class is created.

Exercise

1. Create a **Student class** using default and parameterized constructors to initialize student name and roll number.

Code

```
public class Expriment2 {
    int id;
    String name;

    Expriment2() {
        id = 509;
        name = "Vatsal";
    }

    Expriment2(int i, String n) {
        id = i;
        name = n;
    }

    Expriment2(Expriment2 s) {
        id = s.id;
    }
}
```

```

        name = s.name;
    }

    void display() {
        System.out.println("id:" + id);
        System.out.println("name:" + name);
    }

    public static void main(String[] args) {
        System.out.println("Default Constructor");
        Expriment2 e = new Expriment2();
        e.display();

        System.out.println("Parameterized Constructor");
        Expriment2 e1 = new Expriment2(512, "Parth");
        e1.display();

        System.out.println("Copy Constructor");
        Expriment2 e2 = new Expriment2(e1);
        e2.display();
    }
}

```

Output

```

C:\Users\USER\Downloads\Vatsal\JAVA>javac Expriment2.java

C:\Users\USER\Downloads\Vatsal\JAVA>java Expriment2
Default Constructor
id :509
name :Vatsal
Parameterized Constructor
id :512
name :Parth
Copy Constructor
id :512
name :Parth

C:\Users\USER\Downloads\Vatsal\JAVA>

```

The screenshot shows a Windows command prompt window. The title bar indicates the path 'C:\Users\USER\Downloads\Vatsal\JAVA'. The command prompt shows the following sequence of commands and outputs:

- Command: `javac Expriment2.java`
- Command: `java Expriment2`
- Output:


```

Default Constructor
id :509
name :Vatsal
Parameterized Constructor
id :512
name :Parth
Copy Constructor
id :512
name :Parth

```
- Final command prompt: `C:\Users\USER\Downloads\Vatsal\JAVA>`

The taskbar at the bottom shows the system clock as 18:15 on 17-07-2026, along with various system icons and the language set to ENG.

2. Develop a **Mobile Phone Inventory System** using different constructors to initialize mobile details and create duplicate object records.

Code

```
public class MobileInventory {
    int modelId;
    String brandName;

    public MobileInventory() {
        modelId = 1000;
        brandName = "Iphone";
    }

    public MobileInventory(int id, String brand) {
        modelId = id;
        brandName = brand;
    }

    public MobileInventory(MobileInventory phone) {
        modelId = phone.modelId;
        brandName = phone.brandName;
    }

    void display() {
        System.out.println("Mobile ID : " + modelId);
        System.out.println("Brand Name : " + brandName);
    }

    public static void main(String[] args) {
        System.out.println("Default Constructor");
        MobileInventory Phone1 = new MobileInventory();
        Phone1.display();

        System.out.println("Parameterized Constructor");
        MobileInventory Phone2 = new MobileInventory(2000, "Oppo K13 Turbo");
        Phone2.display();

        System.out.println("Copy Constructor");
        MobileInventory Phone3 = new MobileInventory(Phone2);
        Phone3.display();
    }
}
```

Output

```
C:\Users\USER\Downloads\Vatsal\JAVA>javac MobileInventory.java

C:\Users\USER\Downloads\Vatsal\JAVA>java MobileInventory
Default Constructor
Mobile ID   : 1000
Brand Name  : Iphone
Parameterized Constructor
Mobile ID   : 2000
Brand Name  : Oppo K13 Turbo
Copy Constructor
Mobile ID   : 2000
Brand Name  : Oppo K13 Turbo

C:\Users\USER\Downloads\Vatsal\JAVA>
```

FIFA World Cup 2026... 18:26 17-07-2026 5